

## Section 15.8: Triple Integrals in Spherical Coordinates

Multiple Integrals

MTH 310  
Calculus III

### Why Spherical Coordinates?

Cylindrical coordinates handle solids with circular cross-sections in the  $xy$ -plane. But some solids are more naturally described by their **distance from the origin**:

- Spheres:  $x^2 + y^2 + z^2 = a^2$
- Cones:  $z = c\sqrt{x^2 + y^2}$
- Hemispheres, spherical shells, ice-cream-cone regions

In cylindrical coordinates, a sphere is  $r^2 + z^2 = a^2$  — still messy. In **spherical coordinates**, it becomes simply  $\rho = a$ .

Spherical coordinates describe a point by:

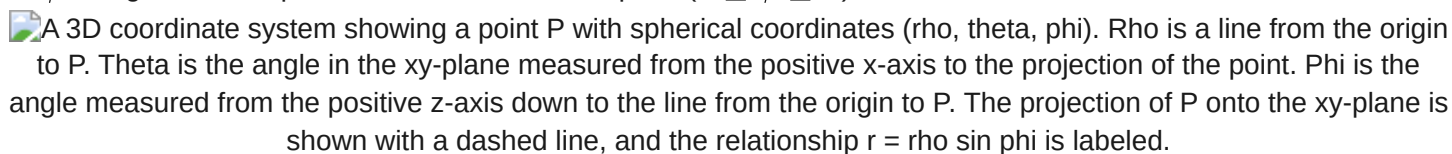
1. How far it is from the **origin** ( $\rho$ )
2. What direction it's in ( $\theta$  and  $\phi$ )

Use spherical when the solid or integrand involves  $x^2 + y^2 + z^2$  (distance from origin). Use cylindrical when it involves  $x^2 + y^2$  (distance from the  $z$ -axis).

### The Spherical Coordinate System

A point in 3D space is described by  $(\rho, \theta, \phi)$  where:

- $\rho$  = distance from the **origin** ( $\rho \geq 0$ )
- $\theta$  = angle in the  $xy$ -plane from the positive  $x$ -axis (same as in cylindrical/polar)
- $\phi$  = angle from the positive  $z$ -axis down to the point ( $0 \leq \phi \leq \pi$ )

A 3D coordinate system showing a point P with spherical coordinates (rho, theta, phi). Rho is a line from the origin to P. Theta is the angle in the xy-plane measured from the positive x-axis to the projection of the point. Phi is the angle measured from the positive z-axis down to the line from the origin to P. The projection of P onto the xy-plane is shown with a dashed line, and the relationship  $r = \rho \sin \phi$  is labeled.

**Important:**  $\phi$  is measured from the **top** (positive  $z$ -axis):

- $\phi = 0$ : pointing straight up (north pole)
- $\phi = \pi/2$ : in the  $xy$ -plane (equator)
- $\phi = \pi$ : pointing straight down (south pole)

$\rho$  tells you how far,  $\theta$  tells you which compass direction,  $\phi$  tells you how far from vertical. Think of it as GPS coordinates on a sphere: longitude ( $\theta$ ) and colatitude ( $\phi$ ).

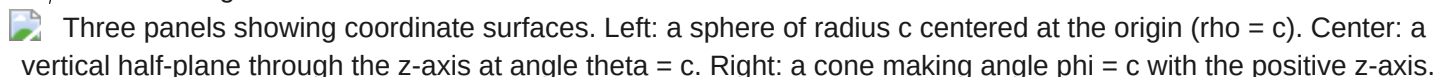
### Coordinate Surfaces: What Does Each Constant Look Like?

What surface do you get when you hold one coordinate constant?

| Equation     | Surface    | Description                          |
|--------------|------------|--------------------------------------|
| $\rho = c$   | Sphere     | Centered at origin, radius $c$       |
| $\theta = c$ | Half-plane | Vertical, hinged on the $z$ -axis    |
| $\phi = c$   | Cone       | Half-cone opening from the $z$ -axis |

Some important special cases for  $\phi$ :

- $\phi = 0$ : the positive  $z$ -axis (a degenerate "cone" — just a ray)
- $\phi = \pi/4$ : a cone at 45 degrees from vertical
- $\phi = \pi/2$ : the  $xy$ -plane (the "widest" cone — a flat disk)
- $\phi = \pi$ : the negative  $z$ -axis

Three panels showing coordinate surfaces. Left: a sphere of radius  $c$  centered at the origin ( $\rho = c$ ). Center: a vertical half-plane through the  $z$ -axis at angle  $\theta = c$ . Right: a cone making angle  $\phi = c$  with the positive  $z$ -axis.

$\rho = \text{const}$  gives a sphere,  $\theta = \text{const}$  gives a half-plane,  $\phi = \text{const}$  gives a cone. These are the building blocks for spherical regions.

## Conversion Formulas

### Spherical $\rightarrow$ Cartesian Conversions

Spherical to Cartesian:

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi$$

Cartesian to Spherical:

$$\rho^2 = x^2 + y^2 + z^2, \quad \tan \theta = \frac{y}{x}, \quad \cos \phi = \frac{z}{\rho}$$

Useful relationship to cylindrical:

$$r = \rho \sin \phi, \quad z = \rho \cos \phi$$

This makes sense: the projection of  $\rho$  onto the  $xy$ -plane is  $\rho \sin \phi$  (the cylindrical  $r$ ), and the vertical component is  $\rho \cos \phi$  (the height  $z$ ).

Common conversions:

| Cartesian              | Spherical            | Surface                  |
|------------------------|----------------------|--------------------------|
| $x^2 + y^2 + z^2 = 16$ | $\rho = 4$           | Sphere of radius 4       |
| $z = \sqrt{x^2 + y^2}$ | $\phi = \pi/4$       | Cone at 45 degrees       |
| $z = 0$                | $\phi = \pi/2$       | The $xy$ -plane          |
| $x^2 + y^2 + z^2 = 2z$ | $\rho = 2 \cos \phi$ | Sphere tangent to origin |

The master identity is  $\rho^2 = x^2 + y^2 + z^2$ . For cones  $z = c\sqrt{x^2 + y^2}$ , divide both sides by  $\rho$  to get  $\cos \phi = c \sin \phi$ , i.e.,  $\phi = \arctan(1/c)$ .

## Quick Check: Convert Equations to Spherical

Write each equation in spherical coordinates:

1.  $x^2 + y^2 + z^2 = 16$

2.  $z = \sqrt{x^2 + y^2}$

For (1), use  $\rho^2 = x^2 + y^2 + z^2$ . For (2), write in terms of  $\rho$ ,  $\phi$  and simplify.

## The Volume Element in Spherical Coordinates

In Cartesian,  $dV = dx dy dz$ . In cylindrical,  $dV = r dz dr d\theta$ . What about spherical?

A small "spherical wedge" at position  $(\rho, \theta, \phi)$  has three edge lengths:

- Radial:  $d\rho$
- Longitudinal arc:  $\rho \sin \phi d\theta$  (circle of radius  $\rho \sin \phi$  in the  $xy$ -plane)
- Latitudinal arc:  $\rho d\phi$  (arc along a meridian)

The volume of this wedge is the product of the three edge lengths:

$$dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

 A small spherical volume element showing a wedge-shaped piece with radial thickness  $d\rho$ , longitudinal arc length  $\rho \sin \phi \, d\theta$ , and latitudinal arc length  $\rho \, d\phi$ . The volume element  $dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$  is labeled.

The Jacobian factor is  $\rho^2 \sin \phi$ . This is the price of using curved coordinates — but it's worth it when the limits become simple constants.

### Quick Check: Describe the Region

What solid does the following region describe?

$$0 \leq \rho \leq 3, \quad 0 \leq \phi \leq \pi/2, \quad 0 \leq \theta \leq 2\pi$$

Think about each limit: what does  $\rho \leq 3$  mean? What does  $\phi \leq \pi/2$  restrict? What does  $\theta$  from 0 to  $2\pi$  mean?

### When to Use Spherical Coordinates

Use spherical when:

1. The region is bounded by **spheres** ( $\rho = a$ ) and/or **cones** ( $\phi = c$ )
2. The integrand involves  $x^2 + y^2 + z^2$  (becomes  $\rho^2$ )
3. The solid is a full ball, hemisphere, spherical shell, or "ice-cream cone" shape

Template:

$$\iiint_E f \, dV = \int_{\alpha}^{\beta} \int_{\phi_1}^{\phi_2} \int_{\rho_1(\theta, \phi)}^{\rho_2(\theta, \phi)} f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

**Typical order:**  $d\rho \, d\phi \, d\theta$ . The  $\rho$ -integral is innermost (simplest limits), and  $\theta$  is outermost (usually 0 to  $2\pi$ ).

The decision tree: **circles in  $xy$ -plane?** → cylindrical. **Spheres or cones from origin?** → spherical. **Neither?** → Cartesian.

## Example 1: Integral Over a Lower Hemisphere

### Example 1

Evaluate  $\iiint_H (9 - x^2 - y^2) dV$ , where  $H$  is the solid lower hemisphere  $x^2 + y^2 + z^2 \leq 9$ ,  $z \leq 0$ .

**Convert everything to spherical:**

- $9 - x^2 - y^2 = 9 - \rho^2 \sin^2 \phi$  (since  $x^2 + y^2 = \rho^2 \sin^2 \phi$ )
- Sphere:  $\rho = 3$
- Lower hemisphere ( $z \leq 0$ ):  $\phi \in [\pi/2, \pi]$

Set up the triple integral with  $dV = \rho^2 \sin \phi d\rho d\phi d\theta$ . What are the limits on  $\rho$ ,  $\phi$ ,  $\theta$ ?

## Example 2: Between Spheres, Above a Cone

### Example 2

Evaluate  $\iiint_E xyz dV$ , where  $E$  lies between the spheres  $\rho = 2$  and  $\rho = 4$ , and above the cone  $\phi = \pi/3$ .

**Identify the region:** A spherical shell ( $2 \leq \rho \leq 4$ ) cut by a cone ( $\phi \leq \pi/3$ ) — like the top of an ice-cream cone.

Convert  $xyz$  to spherical coordinates and set up the integral. Recall  $x = \rho \sin \phi \cos \theta$ ,  $y = \rho \sin \phi \sin \theta$ ,  $z = \rho \cos \phi$ .

## Choosing the Right Coordinate System

After Chapters 15.6–15.8, you have three coordinate systems for triple integrals. How do you choose?

| Coordinate System     | Best When                               | Volume Element  |
|-----------------------|-----------------------------------------|-----------------|
| Cartesian $(x, y, z)$ | Rectangular solids, no obvious symmetry | $dV = dx dy dz$ |

| Coordinate System                       | Best When                                                            | Volume Element                                       |
|-----------------------------------------|----------------------------------------------------------------------|------------------------------------------------------|
| <b>Cylindrical</b> $(r, \theta, z)$     | Circular symmetry about the $z$ -axis: cylinders, paraboloids, cones | $dV = r \, dz \, dr \, d\theta$                      |
| <b>Spherical</b> $(\rho, \theta, \phi)$ | Symmetry about the origin: spheres, cones from origin                | $dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$ |

#### Quick decision guide:

- See  $x^2 + y^2$  in the region or integrand? → **Cylindrical**
- See  $x^2 + y^2 + z^2$  in the region or integrand? → **Spherical**
- Region described by planes only? → **Cartesian**
- Sphere + cone? → **Spherical** (both become constants)
- Cylinder + horizontal planes? → **Cylindrical**

The "right" coordinate system is the one that makes the **limits of integration** simplest. If the boundaries become constants, you've found the right system.

#### Homework

**Section 15.8:** 1, 3, 5, 6, 7, 9, 11, 15, 17, 19, 20, 23, 25, 27, 29, 31, 37, 43, 50